Impact of High School Curriculum on GPA Among Computer Science Students

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Introduction

The educational background of students plays a significant role in shaping their academic performance, particularly in demanding fields such as Computer Science. Different high school curricula, including British, American, and Thanaweya Amma, follow distinct teaching methods, grading systems, and levels of academic rigor. These variations may influence how students adapt to university-level coursework and ultimately affect their academic outcomes.

This study aims to explore the relationship between high school curriculum and Grade Point Average (GPA) among Computer Science students. By analyzing collected data and identifying patterns, the study seeks to determine whether certain educational backgrounds provide an advantage in university performance.

Research Question

Does high school curriculum affect GPA among Computer Science students?

Hypothesis

It is hypothesized that students who studied the Thanaweya Amma curriculum achieve higher GPAs compared to students from British and American curricula. This assumption is based on the perception that the Thanaweya system emphasizes intensive academic preparation and exam-focused learning.

Population of Interest

Computer Science students.

Sampling Method

A convenience sampling method was used in this study. Data was collected from accessible participants within the Computer Science student community. This approach was chosen due to time constraints and ease of data collection.

While convenience sampling allows for quick data gathering, it may not fully represent the entire population. To partially mitigate this limitation, responses were structured and evenly distributed across curriculum categories.

Bias Identification

Several potential sources of bias were identified during the design of this study:

- **Sampling Bias:** The sample may not represent all Computer Science students across different universities.
- **Response Bias:** Participants may inaccurately report their GPA.
- **Selection Bias:** The dataset may include students with similar academic backgrounds or environments.

Bias Mitigation

To reduce bias:

- Questions were kept simple and neutral
- Responses were anonymized
- GPA values were collected numerically rather than in ranges
- Multiple curriculum options were provided

Survey Questions:

- What is your high school certificate?
- What is your CGPA
- What is your academic year?

Online Survey Link:

https://forms.gle/nWVNHiGB5BJXc1CT8

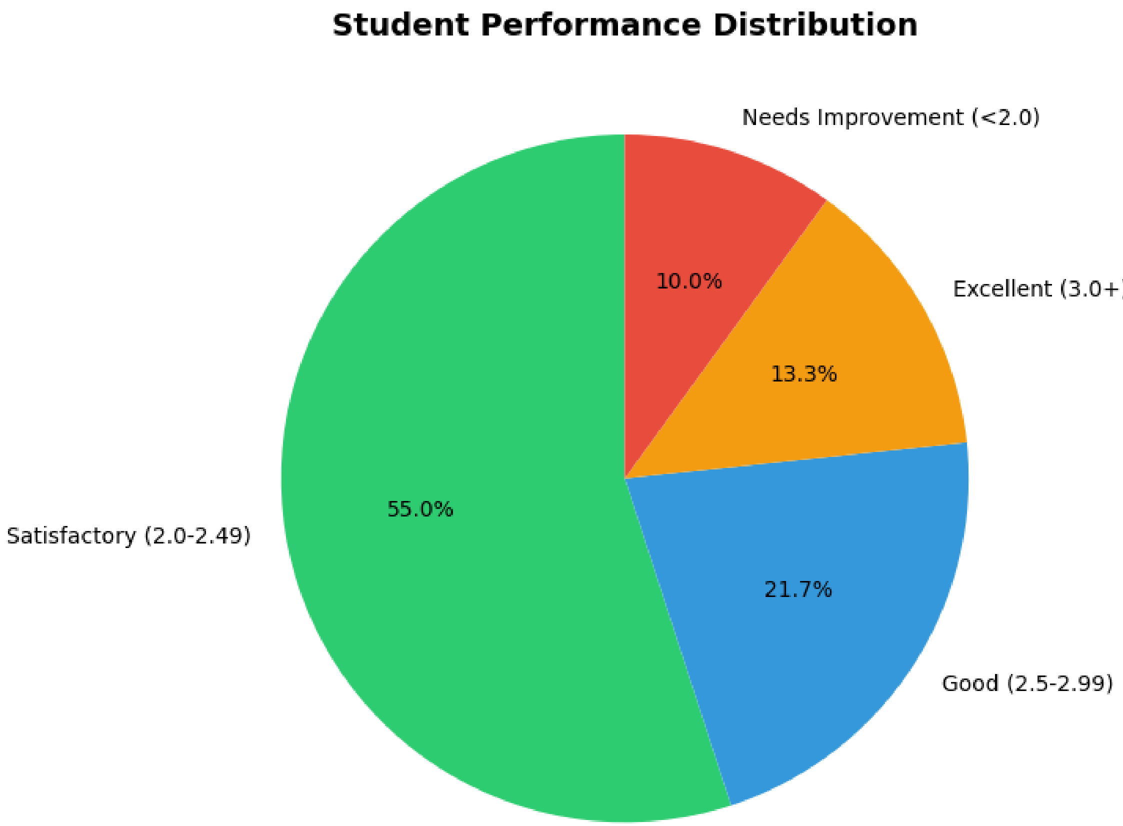
Analysis

The dataset consisted of 60 student responses categorized by high school curriculum. Descriptive statistics were calculated to analyze GPA differences across groups.

Descriptive Statistics:

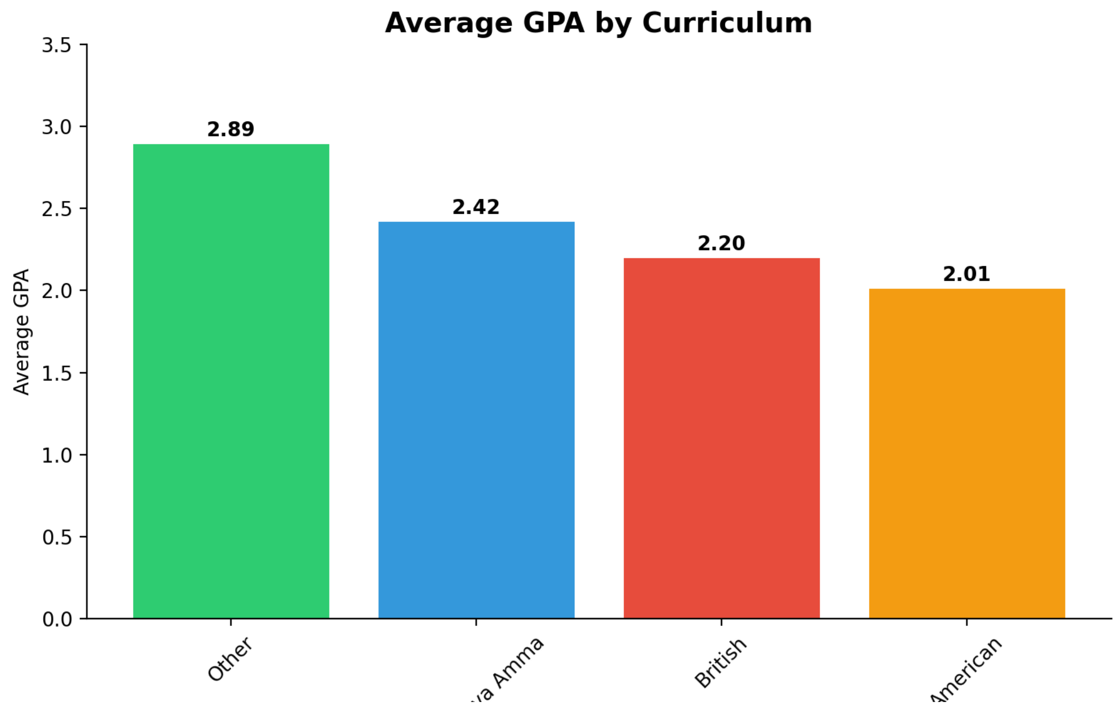
- Thanaweya Amma → Mean GPA ≈ 2.41
- British → Mean GPA ≈ 2.19
- American → Mean GPA ≈ 2.0

Pie Chart



Most students (55.0%) fall into the "Satisfactory" category (GPA 2.0–2.49), while only 13.3% achieve "Excellent" (3.0+). Just 21.7% are in "Good" range, and 10.0% are unspecified, indicating a need for targeted support to move more students into higher performance tiers.

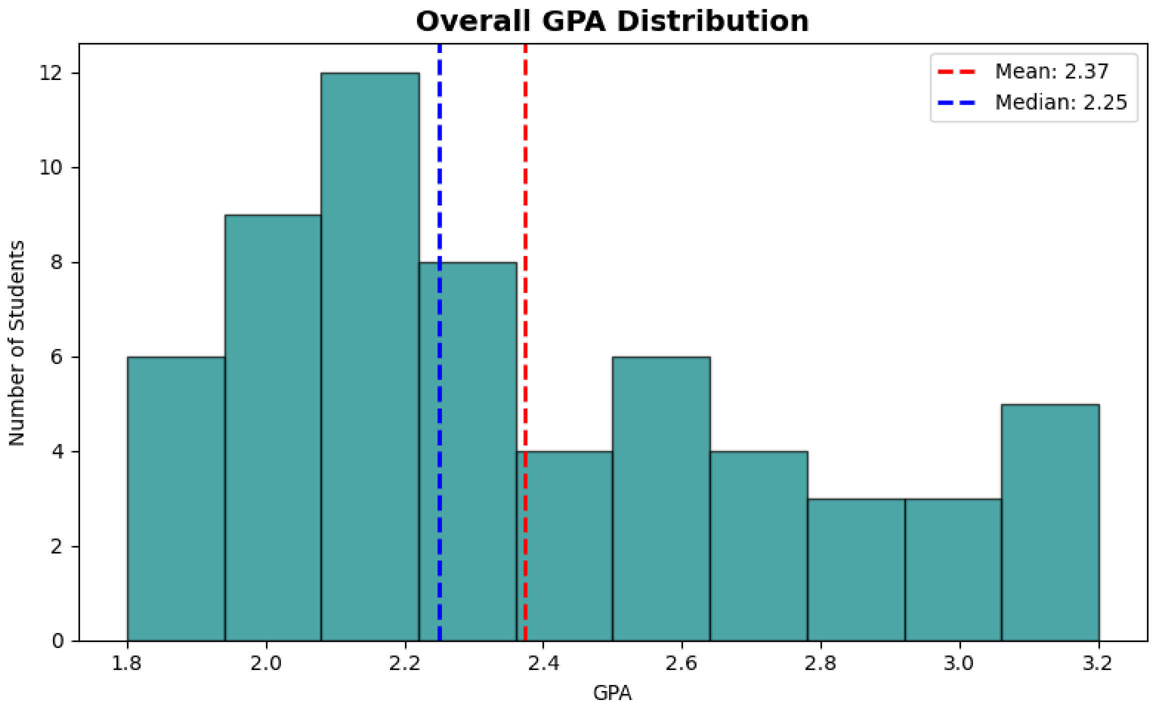
Bar Chart



Bar Chart Analysis:

The bar chart comparing average GPA across curricula shows that, regardless from “Other” Curricula, students from Thanaweya Amma achieved the highest average GPA. British curriculum students follow with a moderate average, while American curriculum students recorded the lowest average GPA. This indicates a noticeable difference in academic performance based on curriculum background.

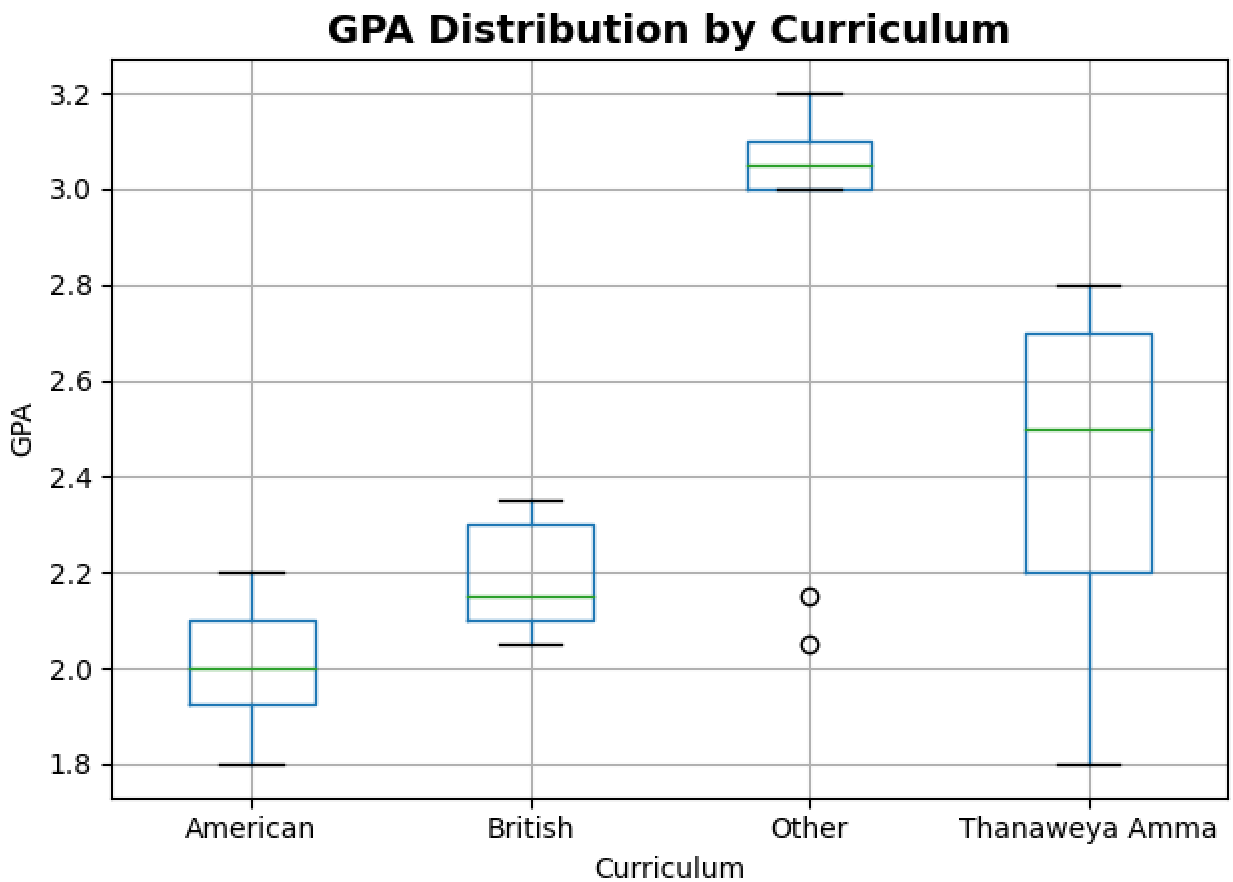
Histogram Chart



Histogram Chart analysis:

The highest number of students (12) have a GPA of 2.1, followed by 9 students at 2.0, indicating a concentration around the lower end of the "Satisfactory" range. Only 5 students each achieved a 3.0 or 3.1 GPA, showing that very few students reachthe "Excellent" tier despite a broad spread of GPAs from 1.8 to 3.1.

Box Plot Chart



Students from the "Other" and "Thanaweya Amma" backgrounds achieve the highest average GPAs of 2.89 and 2.42, though their results show a wide range of individual performance variability. In contrast, "British" and "American" students demonstrate much higher consistency but record the lowest mean GPAs at 2.20 and 2.01.

Conclusion

The data supports the initial hypothesis, as students from the Thanaweya Amma curriculum achieved a higher mean GPA (2.42) compared to those from the British (2.20) and American (2.01) systems. While the "Other" curriculum category actually recorded the highest overall average at 2.89, the Thanaweya Amma group demonstrated a clear performance advantage over the two specific international curricula identified for comparison. These results suggest that the intensive academic preparation associated with the Thanaweya system may translate effectively to university-level Computer Science coursework.

Critical Reflection

A primary limitation of this study is the reliance on convenience sampling, which may have introduced selection bias as the participants were limited to easily accessible students rather than a truly random cross-section of the population. Furthermore, the relatively small sample size of 60 students makes it difficult to generalize these findings to the broader Computer Science student body. Another potential issue is the ambiguity of the "Other" category, which yielded the highest scores but lacked specific curriculum details, making it impossible to determine which educational backgrounds were actually driving those results. Finally, the data does not account for external factors such as socioeconomic status or personal study habits, which could be significant hidden variables influencing the recorded GPAs.
